Standalone Multi-Platform Scraper API

FRONTEND INTEGRATION GUIDE

Supported platforms

Instagram | Facebook | X | YouTube

API base URL

```
http://localhost:4000
```

Purpose

A complete handoff for integrating source/profile submission, asynchronous scrape jobs, polling, result pagination, cancellation, deletion, optional comments, OCR, transcription, and completion callbacks.

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1. Integration at a glance

Critical frontend security rule
Do not place API_KEY in browser JavaScript, mobile bundles, or public environment variables. The current service has no CORS middleware. A browser frontend should call its own backend/BFF or a same-origin reverse proxy, which then calls this scraper API with X-API-Key.

Recommended request flow

Step	Frontend / BFF action	Expected result
1	Submit POST /v1/scrape-jobs	HTTP 202 with job ID and URLs
2	Store the returned job ID	Use it as the stable UI correlation key
3	Poll GET /v1/scrape-jobs/:id	Render queued/running/terminal progress
4	Stop polling at a terminal status	completed, partially_completed, failed, or cancelled
5	Fetch GET /v1/scrape-jobs/:id/results	Render profiles, posts, comments, media, enrichment, and warnings
6	Paginate results	Increment page until fewer than pageSize items are returned

Service topology

Component	Location	Frontend relevance
API	Host process, port 4000	Only HTTP surface the BFF calls
Worker	Host process	Processes queued jobs; no direct frontend access
PostgreSQL	Docker, localhost:15433	Persistence; frontend must not connect directly
Redis	Docker, localhost:6380	BullMQ transport; frontend must not connect directly
Drizzle Studio	http://localhost:4983	Developer inspection only

Authentication

Every endpoint under **/v1** requires the request header below. **/health** and **/ready** are public.

```
X-API-Key: <API_KEY>
```

HTTP	Meaning	Frontend handling
401	Missing or invalid X-API-Key	Do not retry; fix BFF/server configuration
400	Schema/query validation failed	Show actionable validation feedback
404	Job or content was not found	Treat stale IDs as unavailable
422	Feature disabled or enrichment storage missing	Disable unsupported UI option or correct server configuration
500	Unexpected server/provider/storage error	Show retry option and log sanitized error
503	Database or Redis not ready	Retry readiness with backoff

2. Create a scrape job

POST /v1/scrape-jobs

Creates or reuses a source, upserts one feed per submitted platform, persists a queued job, and schedules asynchronous work. Returns HTTP 202.

Field	Type	Required / default	Rules and meaning
source	object	Required	Either {name, description?} for a new source OR {id} for an existing source UUID.
feeds	array	Required	1-4 entries. Each entry has url and optional platform. At most one feed per platform.
feeds[].platform	enum	Auto-detected	instagram, facebook, x, youtube. Pass explicitly when detection may be ambiguous.
startDate	ISO datetime	See date rules	Inclusive provider lower boundary. Must be earlier than endDate when both supplied.
endDate	ISO datetime	See date rules	Stored as requested end boundary; clients should always send both boundaries.
postLimit	integer	100	Range 1-200; provider maximum target, not a guaranteed count.
includeComments	boolean	false	When true, schedules comments for posts reporting commentsCount > 0.
commentLimit	integer	100	Range 1-100 per post.
skipPinnedPosts	boolean	true	Instagram pinned posts are skipped by default.
maxVideoDurationSeconds	integer	180	Positive, maximum 86400. Posts over the cap are excluded.
ocr	boolean	false	Requires OCR_ENABLED=true plus S3_BUCKET and S3_PUBLIC_BASE_URL.
transcription	boolean	false	Requires TRANSCRIPTION_ENABLED=true plus S3 settings.
callbackUrl	HTTPS URL	Optional	Receives a signed completion callback. HTTP URLs are rejected.

Date-window rule
If both startDate and endDate are omitted, the server resolves a fixed 24-hour UTC window ending at request time. If either is supplied, the default is disabled. Frontend integrations should always send both ISO-8601 UTC values to avoid an open boundary.

New source example

```
POST /v1/scrape-jobs
Content-Type: application/json
X-API-Key: <server-side secret>

{
"source": { "name": "Sara Gonzales" },
"feeds": [
{ "platform": "instagram", "url": "https://www.instagram.com/saragonzalestx/" },
{ "platform": "facebook", "url": "https://www.facebook.com/SaraGonzalesTX/" }
],
"startDate": "2026-10-03T00:00:00.000Z",
"endDate": "2026-10-06T00:00:00.000Z",
"postLimit": 100,
"includeComments": false,
"ocr": false,
"transcription": false
}
```

Existing source example

```
{
"source": { "id": "1f0569eb-733d-4993-bd8c-8b52e31c4d7b" },
"feeds": [
{ "platform": "instagram", "url": "https://instagram.com/example" }
]
}
```

Create response

```
HTTP/1.1 202 Accepted
{
"id": "5757ff7b-f6a5-4cf9-ae99-8aa52dcc9e2d",
"status": "queued",
"statusUrl": "http://localhost:4000/v1/scrape-jobs/5757ff7b-f6a5-4cf9-ae99-8aa52dcc9e2d",
"resultsUrl": "http://localhost:4000/v1/scrape-jobs/5757ff7b-f6a5-4cf9-ae99-8aa52dcc9e2d/results"
}
```

Platform URL detection

Platform	Recognized URL/value
Instagram	Host contains instagram.com
Facebook	facebook.com or fb.com
X	x.com or twitter.com
YouTube	youtube.com, youtu.be, or a channel ID beginning UC

Important source/feed behavior

• Creating with source.name always creates a new source row; use source.id to add or refresh feeds under an existing source.

• One source can have exactly one Instagram, Facebook, X, and YouTube feed. Resubmitting the same platform for an existing source updates that feed URL.

• Profile and post scraping are always attempted. Comments and enrichment are opt-in.

• A successful job may legitimately contain zero posts when no posts match the requested date window or platform filters.

3. Read job status

GET /v1/scrape-jobs/:id

Poll this endpoint while the job is non-terminal. A 2-5 second interval is appropriate for an active screen; back off for background tracking.

Job status	Terminal?	Meaning
queued	No	Accepted and waiting for orchestration
running	No	At least one target is active
completed	Yes	All targets completed without recorded target errors
partially_completed	Yes	Usable data exists, but at least one target or optional stage recorded an error
failed	Yes	No target produced usable completion
cancelled	Yes	Cancellation was requested

Status response shape

```
{
"id": "<job UUID>",
"source_id": "<source UUID>",
"status": "running",
"request": { "...": "original resolved request" },
"callback_url": null,
"progress": {
"totalTargets": 2,
"completedTargets": 1,
"successfulTargets": 1,
"failedTargets": 0
},
"error": null,
"cancel_requested": false,
"created_at": "<ISO timestamp>",
"started_at": "<ISO timestamp or null>",
"completed_at": null,
"expires_at": "<ISO timestamp>",
"targets": [
{
"id": "<target UUID>",
"feed_id": "<feed UUID>",
"input_url": "https://...",
"platform": "instagram",
"status": "completed",
"pending_tasks": 0,
"error": null
}
]
}
```

UI interpretation
Do not treat completedTargets as successfulTargets. A partially completed target counts as completed terminal work and also as usable success. Display target.error and later result warnings without exposing them as raw stack traces.

4. Fetch normalized results

GET /v1/scrape-jobs/:id/results?page=1&pageSize;=50

Query	Default	Allowed	Notes
page	1	Positive integer	1-based
pageSize	50	1-100	Applied independently to contents, comments, media, and enrichment arrays

Pagination caveat
The response does not currently include total counts or next-page tokens. Continue requesting pages until every paginated array you need returns fewer than pageSize entries. source and feeds repeat on each page.

Top-level result object

Property	Shape	Meaning
page, pageSize	number	Echoed pagination values
source	object | null	The logical person/organization source
feeds	array	Normalized platform profiles attached to the source
contents	array	Posts/videos/images/text ordered newest first
comments	array	Optional normalized comments
media	array	Original and optional durable media metadata
enrichment	array	Caption, precheck, transcription, and OCR results
warnings	array	Failed provider attempts only

Feed/profile fields

Field	Use in UI
id, source_id	Stable database identifiers
platform, url	Platform identity and submitted profile URL
url_verification_status	pending, pending_retry, completed, or failed
name, username, about	Normalized profile identity
external_id	Platform-specific profile ID
profile_picture	Provider profile image URL
followers_count, follows_count, content_count	64-bit database values; JSON may serialize them as strings
is_verified_account	boolean or null
last_updated_at	Last successful profile refresh
profile_raw	Provider payload retained for debugging; avoid coupling UI to it

Content/post fields

Field group	Fields
Identity	id, job_id, target_id, source_id, feed_id, platform, external_id, url
Display	type, title, description, raw_title, raw_description, thumbnail_url, hashtags
Timing/media	published_at, duration_seconds
Metrics	likes_count, comments_count, shares_count, views_count, engagement_rate
Derived/reserved	transcript, OCR text, summary, sentiment, categories, district, language, political/governance flags, stance, actors, claim, reasoning
Debug	provider_raw, created_at, updated_at

Numeric handling
PostgreSQL bigint and numeric fields may be returned by the Node PostgreSQL driver as strings. Frontend types should accept string | number | null for counts and duration, then parse deliberately. Avoid implicit arithmetic on raw JSON values.

Result response example


```
{
"page": 1,
"pageSize": 50,
"source": { "id": "<uuid>", "name": "Sara Gonzales" },
"feeds": [
{
"platform": "facebook",
"name": "Sara Gonzales",
"username": "SaraGonzalesTX",
"followers_count": "466000",
"url_verification_status": "completed"
}
],
"contents": [
{
"id": "<uuid>",
"platform": "facebook",
"type": "video",
"external_id": "1679517113532544",
"published_at": "2026-10-06T01:00:26Z",
"url": "https://www.facebook.com/reel/.../",
"likes_count": "0",
"comments_count": "0"
}
],
"comments": [],
"media": [
{
"content_id": "<uuid>",
"media_type": "video",
"original_url": "https://...",
"durable_url": null,
"mime_type": null,
"status": "original"
}
],
"enrichment": [],
"warnings": []
}
```


Warnings

Each warning contains provider, capability, error, and created_at. The API exposes only failed provider attempts here; successful attempts remain operational database records.

5. Cancel and delete jobs

POST /v1/scrape-jobs/:id/cancel

Requests cancellation, marks queued/running work cancelled, and stops scheduling later stages where workers honor cancel_requested. In-flight external requests may finish.

```
HTTP/1.1 202 Accepted
{
"id": "<job UUID>",
"status": "cancelled",
"cancelRequested": true
}
```

DELETE /v1/scrape-jobs/:id

Deletes the scrape job and cascaded job-owned results. If durable scraper-owned media exists, object deletion is queued. Returns **204 No Content**. This is destructive and should require UI confirmation.

Condition	Cancel	Delete
Unknown ID	404 {error: job not found}	404 {error: job not found}
Success	202 JSON response	204 empty response
Recommended UI	Allow while queued/running	Confirm destructive action; remove local cache

6. Health and readiness

Endpoint	Auth	Success	Failure
GET /health	None	200 {status: ok}	Process unreachable
GET /ready	None	200 {status: ready, postgres: ok, redis: ok}	503 {status: not_ready}

Use /ready for deployment probes and before accepting user submissions. /health only confirms that the HTTP process responds.

7. Comments, OCR, captions, and transcription

Capability	Job flag	Global requirement	Processing rule
Comments	includeComments	None	Only posts reporting commentsCount > 0 are scheduled; commentLimit is per post.
OCR	ocr	OCR_ENABLED=true plus S3	Only image content is copied and sent to OCR.
Transcription	transcription	TRANSCRIPTION_ENABLED=true plus S3	Eligible video/audio is copied, prechecked, then dispatched when approved.
YouTube captions	Automatic	YouTube/Media credentials	Captions are preferred. Missing captions may route to transcription if requested/enabled.

Enrichment create-time errors

Condition	Response
ocr=true while OCR_ENABLED=false	422 {error: OCR is globally disabled}
transcription=true while TRANSCRIPTION_ENABLED=false	422 {error: Transcription is globally disabled}
OCR/transcription requested without S3 bucket/public URL	422 {error: S3_BUCKET and S3_PUBLIC_BASE_URL are required for enrichment}

8. Service webhook APIs

These are service-to-service callbacks, not ordinary frontend endpoints:

• POST /v1/webhooks/video-precheck

• POST /v1/webhooks/transcription

• POST /v1/webhooks/ocr

Required headers

```
X-API-Key: <API_KEY>
X-Webhook-Timestamp: <Unix seconds>
X-Webhook-Signature: <lowercase hex HMAC-SHA256>
```

Signature input

```
signature = hex(
HMAC_SHA256(
ENRICHMENT_WEBHOOK_SECRET,
timestamp + "." + rawJsonBody
)
)
```

The raw JSON bytes must match the bytes used to calculate the signature. Timestamps older/newer than five minutes are rejected. All three endpoints are also behind X-API-Key.

Webhook body

```
{
"requestId": "external-idempotency-key",
"postId": "<content UUID>",
"status": "completed",
"text": "optional recognized text",
"result": { "optional": "service-specific object" },
"error": "optional failure message"
}
```

Field	Requirement
requestId	Required non-empty string; replay/idempotency key
postId	Required UUID of contents.id
status	completed or failed
text	Optional OCR/transcription/caption text
result	Optional JSON object; precheck may include should_transcribe, transcriptMediaUrl, and language
error	Optional failure description

Responses

Response	Meaning
202 {accepted: true}	Accepted and persisted
202 {accepted: true, duplicate: true}	requestId was already processed
401 {error: invalid webhook signature}	Missing, stale, or invalid signature
404 {error: post not found}	postId does not exist

9. Completion callbacks to your application

If callbackUrl is included when creating a job, the scraper posts a completion event after a terminal outcome. The URL must use HTTPS.

Callback body

```
{
"id": "<job UUID>",
"status": "completed",
"progress": {
"totalTargets": 2,
"completedTargets": 2,
"successfulTargets": 2,
"failedTargets": 0
},
"resultsUrl": "http://localhost:4000/v1/scrape-jobs/<id>/results"
}
```

Callback headers

```
X-Scraper-Timestamp: <Unix seconds>
X-Scraper-Signature: hex(HMAC-SHA256(
CALLBACK_SIGNING_SECRET,
timestamp + "." + rawJsonBody
))
```

Receiver requirements
Verify the signature against the raw request body, enforce a short timestamp tolerance, process idempotently by job ID, and return 2xx quickly. Fetch final results separately using authenticated server-side API access.

10. Frontend/BFF reference implementation

Server-side TypeScript client

```
const SCRAPER_BASE_URL = process.env.SCRAPER_BASE_URL!;
const SCRAPER_API_KEY = process.env.SCRAPER_API_KEY!;

async function scraperFetch(path: string, init: RequestInit = {}) {
const response = await fetch(`${SCRAPER_BASE_URL}${path}`, {
...init,
headers: {
"content-type": "application/json",
"x-api-key": SCRAPER_API_KEY,
...init.headers,
},
});
if (!response.ok) {
const body = await response.json().catch(() => ({}));
throw new Error(body.error ?? `Scraper HTTP ${response.status}`);
}
return response.status === 204 ? null : response.json();
}

export function createScrapeJob(input: unknown) {
return scraperFetch("/v1/scrape-jobs", {
method: "POST",
body: JSON.stringify(input),
});
}

export function getScrapeJob(id: string) {
return scraperFetch(`/v1/scrape-jobs/${encodeURIComponent(id)}`);
}

export function getScrapeResults(id: string, page = 1, pageSize = 50) {
return scraperFetch(
`/v1/scrape-jobs/${encodeURIComponent(id)}/results?page=${page}&pageSize=${pageSize}`
);
}
```

Polling outline

```
const TERMINAL = new Set([
"completed", "partially_completed", "failed", "cancelled"
]);

async function waitForJob(id: string, signal?: AbortSignal) {
while (!signal?.aborted) {
const job = await getScrapeJob(id);
if (TERMINAL.has(job.status)) return job;
await new Promise(resolve => setTimeout(resolve, 3000));
}
throw new DOMException("Polling cancelled", "AbortError");
}
```

Suggested UI states

API state	Recommended UI
queued	Submitted / waiting
running + target running	Scraping profiles/posts
running + target enriching	Processing comments/media/enrichment
completed	Success; load results
partially_completed	Show available results plus warning banner
failed	Show sanitized target errors and retry action
cancelled	Stopped by user/system

11. Platform behavior and limits

Platform	Primary	Fallback	Built-in filters
Instagram	Local Botsetgo-compatible API	Apify Instagram	Pinned posts skipped by default; duration cap
Facebook	Local Facebook scraper API	Apify Facebook	Duration cap
X	X API v2	None	Replies and retweets excluded; duration cap
YouTube	YouTube Data API	Caption path can use Media API/Apify	Live content excluded; duration cap; captions first

Apify rollover

Controlled globally by **APIFY_FALLBACK_ENABLED**. When false, Instagram and Facebook stay on their local APIs and YouTube captions stay on the configured Media API. A 400, 401, 403, or 422 provider response is treated as non-fallback-eligible; transient failures may roll over.

Data retention

Jobs expire after RETENTION_DAYS (30 by default). A daily worker cleanup removes expired database records and scraper-owned S3 objects. The frontend should not treat the scraper as permanent archival storage unless retention is changed.

12. Integration checklist

- ■ Route browser traffic through a trusted backend/BFF; never expose API_KEY.
- ■ Configure SCRAPER_BASE_URL and API key as server-only secrets.
- ■ Send both startDate and endDate as ISO-8601 UTC strings.
- ■ Allow at most one feed per platform in the create-job form.
- ■ Treat create as asynchronous and persist the returned job ID.
- ■ Poll status until a terminal state or consume the signed completion callback.
- ■ Handle partially_completed as usable data plus warnings.
- ■ Accept bigint/numeric fields as strings in frontend types.
- ■ Paginate results until the requested arrays return fewer than pageSize items.
- ■ Require confirmation before DELETE; stop polling after cancellation.
- ■ Expose OCR/transcription toggles only when server capabilities are enabled.
- ■ Render zero posts as a valid empty date-window result, not automatically as an error.
- ■ Do not build UI logic against profile_raw or provider_raw payload shapes.
- ■ Plan for 30-day retention or export results to the application database.

Local developer references

Item	Value
API	http://localhost:4000
Health	http://localhost:4000/health
Readiness	http://localhost:4000/ready
Drizzle Studio	http://localhost:4983
PostgreSQL	localhost:15433 / database scraper
Redis	localhost:6380

End of guide